

CHAPTER – II

IDEATION PHASE

1.3 Problem Statement:

Heart disease is one of the most serious health issues affecting millions of people worldwide each year, and the number of cases continues to rise, making it a major concern for healthcare providers. Hospitals and healthcare organizations generate and store vast amounts of patient data, which can be difficult to analyse manually. Various factors, including age, gender, blood pressure, cholesterol levels, chest pain type, ECG results, and lifestyle habits, contribute to the development of heart disease. Understanding the relationship between these factors and heart disease becomes challenging when dealing with raw and complex datasets. Traditional methods of data analysis, such as spreadsheets and reports, are often time-consuming and may fail to provide clear insights.

As a result, important trends and patterns can remain hidden within large volumes of data, making it difficult to identify high-risk patients. Effective identification of individuals who are more susceptible to heart disease requires the analysis of multiple factors simultaneously. Doctors and healthcare professionals need accurate, easy-to-understand information to support clinical decision-making, while hospital administrators require clear insights into disease trends for better planning and resource management. Additionally, patients may struggle to understand medical reports due to complex terminology and numerical data. With the continuous growth of healthcare data, there is an increasing need for efficient methods to organize, analyse, and interpret information. Therefore, identifying key risk factors and disease patterns through data visualization and analytical tools is essential for improving patient care, supporting preventive strategies, and enhancing overall healthcare outcomes.